

Computational Neuroscience — Module 08: Planning

— Study Questions

1. What is the default mode network (DMN)? Name its key regions.
2. When is the DMN most active? What cognitive functions does it support?
3. How does the DMN relate to Active Inference's concept of "offline" model simulation?
4. How does the hippocampus construct future scenarios? Why is this called "constructive memory"?
5. What happens to imagination in patients with hippocampal damage?
6. Why do memory and imagination share neural machinery?
7. What is the goal-directed system? Which brain structures implement it?
8. What is the habitual system? Which brain structures implement it?
9. How does Active Inference explain the transition from goal-directed to habitual behavior?
10. What are time cells? How do they represent temporal sequences?
11. How do place cells and time cells combine to form a space-time map?
12. What is the difference between the caudate nucleus and the putamen in action selection?
13. How does frontal lobe damage affect planning ability?
14. How does depression relate to DMN dysfunction?
15. How does addiction disrupt the balance between goal-directed and habitual behavior?
16. What is mind-wandering? Is it useful or wasteful from an Active Inference perspective?
17. How does the brain decide between a familiar habit and a novel plan?
18. What neural evidence supports the idea that imagining the future is "mental time travel"?
19. How does working memory (PFC) interact with episodic memory (hippocampus) during planning?
20. Looking back at the entire Computational Neuroscience unit, how do all eight modules (from systems to planning) show the brain implementing Active Inference at every level?